University of Pennsylvania
CIS 5680: Game Design Practicum

Game Design Document

Umbra

A Light and Shadow Puzzle Game

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1 Executive Summary

Umbra is a top-down 2D spatial puzzle game built in Unreal Engine 5 for PC (Windows). Shadows are rendered continuously and update in real time as objects in the scene move: every light source and blocking object casts live shadow geometry that physically affects the scene. Players manipulate portable lanterns and pivotable spotlights—dragging and rotating them freely with the mouse within designer-specified boundaries—to reshape the shadow pattern across the environment. Blocking objects (crates, pillars) slide along fixed trajectories to designer-specified stop positions when the player interacts with them. Shadow regions are physically solid—a gap becomes crossable only when a shadow-bridge is cast across it, and displacing that bridge destroys the path. Each level is a self-contained puzzle with a correct configuration of light sources and blocking objects that opens a walkable path from start to exit.

Umbra targets fans of clean, mechanic-first puzzle games such as *The Witness*, *Portal*, and *Stephen's Sausage Roll*. There is no combat, no time pressure, and no randomness—every solution is deterministic and discoverable through spatial reasoning alone. A single foundational rule—shadows are physically solid—combines with freely positionable light sources and trajectory-constrained blocking objects to produce a surprisingly deep puzzle space without requiring the player to learn a large system. The game is accessible to players aged 12 and up and designed for pick-up-and-play sessions of 5–15 minutes per level.

Development is structured in three phases across a three-person team: an *Alpha* prototype delivering one fully playable level with the core shadow mechanic (due Wed. Mar. 18); a *Beta* expanding to multiple puzzle levels with polished art, audio, and a hint system (due Wed. Mar. 25); and a *Final* release incorporating a tutorial level, level-select screen, and gameplay/trailer videos (due Mon. Mar. 30 and Wed. Apr. 1). The full prototype is expected to be completed in approximately five weeks of part-time development.

2 Game Design

2.1 High Concept

In **Umbra**, shadows are not merely visual—they are solid, traversable terrain. Players manipulate portable lanterns and pivotable spotlights by dragging and rotating them with the mouse to cast shadows across the scene. Any region the shadow pattern covers becomes physically solid ground; lit regions are open space. Blocking objects (crates, pillars) can be slid along fixed trajectories to redirect or reshape shadow geometry. Each level is a self-contained spatial puzzle: find the correct positions and orientations of all light sources and the correct positions of all blocking objects to open a continuous shadow path from start to exit.

The game is a top-down 2D puzzle built in Unreal Engine 5 for PC (Windows). There is no combat, no time pressure, and no randomness. The entire puzzle space emerges from one rule: *shadows are solid*.

2.2 Design Goals

2.2.1 Main Design Features

Player Goals and Objectives The player is an explorer navigating a world where light and shadow are as solid as stone. The goal of each self-contained level is to discover the correct positions and orientations of light sources and the correct positions of blocking objects along their fixed trajectories, casting a shadow pattern that opens a walkable path from start to exit. There is no time pressure and no combat; the sole challenge is spatial reasoning.

Main Rules and Procedures

- Shadows are continuous and computed in real time. Any region covered by a shadow pattern is physically solid and traversable; lit regions are open space. Bridges, platforms, and doorways built from shadow geometry appear or vanish as the shadow pattern changes.
- **Light sources** (lanterns, spotlights) are freely moved or rotated by clicking and dragging with the mouse. Each light source has a designer-specified boundary that limits how far it can travel or rotate. Shadow geometry updates continuously as the player drags.
- **Blocking objects** (crates, pillars) slide along fixed, pre-defined trajectories and snap to designer-specified stop positions when the player moves to specific pressure plates. They cannot be freely placed; each object has exactly the set of positions its trajectory permits.
- The level is solved when the player's character reaches the exit through a walkable path that exists in the current shadow configuration.

Player Resources

- **Portable lanterns.** Freely dragged to any position within their movement boundary by clicking and holding with the mouse. Cast a circular point-light shadow pattern that updates in real time as the lantern is repositioned.
- **Wall-mounted spotlights.** Freely rotated about their fixed mount point by clicking and dragging. Emit a directed cone of light; rotating the spotlight continuously reshapes the shadow it casts behind any blocking object in its beam.
- **Blocking objects** (crates, pillars). Player moves to pressure plates to toggle position changes of the object to other set positions. Reposition them to add, remove, or redirect a shadow cast by a nearby light source.

Conflicts

- **Spatial logic.** The light source positions and blocking object stop positions that solve the puzzle are hidden; players must deduce the correct configuration through observation and reasoning rather than trial and error.
- **Shadow interdependence.** Moving a light source or blocking object continuously reshapes the entire shadow pattern. A position that creates one bridge may simultaneously remove another the player needs. Players must reason about the full shadow layout, not just local effects.
- **Trajectory constraints.** Blocking objects travel only along their fixed paths and stop only at designer-specified positions. The puzzle solution must be achievable within those constraints; players cannot place objects arbitrarily.

2.2.2 Appeal

Umbra targets fans of pure, mechanic-first puzzle games—players aged 12 and up who enjoy titles such as *The Witness* and *Portal*. A single elegant rule (shadows are solid) produces a deep puzzle space accessible to newcomers yet satisfying for veterans. Every level has a clean "aha" moment when the correct light arrangement snaps into place. No combat, timers, or randomness—only the satisfaction of spatial reasoning.

2.2.3 Look and Feel

Umbra uses a **minimalist silhouette aesthetic** with high contrast between lit and shadowed regions. The live shadow boundary shifts continuously as the player manipulates light sources or blocking objects, making cause and effect immediately visible. Objects are bold geometric silhouettes with no decorative detail; light sources are visually distinct from passive blockers. Ambient audio is calm and atmospheric, reinforced by crisp sound cues for each interaction.

2.3 Worlds, Characters and Story

2.3.1 Back Story

An ancient Roman civilization once mastered the art of sculpting light—using mirrors, lenses, and sacred flames to build bridges and open sealed vaults across their floating island temples. When the empire fell, the islands drifted apart and the light paths collapsed. The player is an explorer who has rediscovered one of these ruined sanctuaries, and must use the lanterns and spotlights left behind to reconstruct the shadow geometry that once connected the fragments and reach the vault at the heart of the ruins.

2.3.2 Spaces / Worlds

All levels are set on the ruins of floating Roman island temples suspended above a void. Each level is a single self-contained island platform viewed from a fixed top-down perspective. The game ships with four spaces: one tutorial and three puzzle levels.

- **Tutorial — The Antechamber.** A small, open platform with one lantern and one blocking pillar. Introduces the core mechanic: the player drags the lantern to cast a shadow across a gap, walks across it, then slides the pillar to extend the shadow to the exit. No dead ends; the correct configuration is reached through minimal experimentation.

- **Level 1 — The Collapsed Bridge.** A medium-sized island with a wide central chasm. One lantern and two blocking pillars are available. The player must position the lantern and arrange the pillars to cast overlapping shadows that span the gap. Introduces the idea that multiple objects must be configured together.

- **Level 2 — The Torch Hall.** A longer, corridor-like island with two wall-mounted spotlights and two blocking objects. The spotlights must be rotated to project shadows down branching side paths; only one branch leads to the exit. Introduces spotlight rotation and the need to reason about shadow direction.

- **Level 3 — The Inner Sanctum.** The largest island, combining a lantern, two spotlights, and three blocking objects. The path to the exit requires using all light sources in concert; adjusting one disrupts another. Intended as the capstone challenge that exercises all mechanics introduced in earlier levels.

2.3.3 Characters

The sole character is **the Orb**, a Roman ceremonial stone sphere that once served as the focal lens at the heart of each island temple. When the empire fell, the Orb was scattered across the ruins and lost its light-guiding purpose. The player controls the Orb, rolling it through the reconstructed shadow paths to return it to the vault at the level's exit. There are no other characters, enemies, or NPCs; the Orb navigates the ruins alone.

2.3.4 Levels of Difficulty

Difficulty scales across the four levels along three axes: the number of interactable objects, the number of mechanics in play simultaneously, and the degree of shadow interdependence (how much moving one object disrupts the rest of the configuration).

- **Tutorial.** One lantern, one pillar, one mechanic (lantern drag). The solution is reachable through direct experimentation; no interdependence between objects.
- **Level 1.** One lantern, two pillars. Both lantern positioning and pillar placement must be correct simultaneously, introducing mild interdependence.
- **Level 2.** Two spotlights, two blocking objects; no lantern. Introduces spotlight rotation as a second mechanic. The branching layout means shadow direction determines which paths are accessible, requiring directional reasoning.
- **Level 3.** One lantern, two spotlights, three blocking objects. All mechanics active at once; high interdependence means most single adjustments require re-evaluating the full shadow layout.

2.4 Interaction Models

2.4.1 User Interface — Navigation and Movement

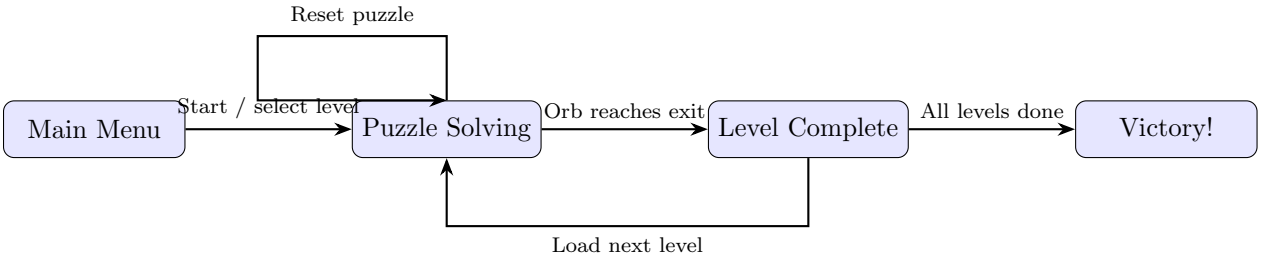
The game uses a fixed top-down camera with no scrolling; the entire level is visible at once. The player uses two input modes simultaneously:

- **Keyboard (WASD / arrow keys).** Moves the Orb through the scene. The Orb is able to traverse shadow-covered regions; entering a lit gap causes it to fall and respawn at the level start position with all objects remaining where the player left them.
- **Mouse.** Used exclusively to manipulate light sources and blocking objects. Hovering over an interactable object highlights it; clicking and dragging repositions lanterns (within their boundary), rotates spotlights (about their fixed mount), or slides blocking objects to the next stop along their trajectory.

The HUD is minimal: the level name is displayed in a corner and a single **Reset** button returns all objects to their starting configuration and the Orb to the level start. There is no timer, health bar, or inventory.

2.4.2 Game Play Sequence and Levels

Each session begins at the Main Menu, where the player selects a level. Within a level the player alternates between configuring light sources and blocking objects and moving the Orb. When the Orb reaches the exit gate the level is marked complete and the next level loads. Completing all three puzzle levels triggers the Victory screen. The player may reset the current level's object configuration at any time without penalty.



2.4.3 User/Environment — Obstacles and Props

- **Gaps / void.** Any region not covered by a shadow pattern is open void. The Orb falls if it enters a lit gap and respawns at the level start; no objects are moved by the respawn.
- **Static ruins geometry.** Walls, archways, and raised platforms are permanently solid and block both the Orb and shadow propagation. They cannot be moved.
- **Portable lanterns.** Interactable light sources; dragged freely within their boundary. Cast omni-directional shadow behind any object they illuminate.
- **Wall-mounted spotlights.** Interactable light sources; rotated about their fixed mount point. Cast a directed cone of shadow behind objects in their beam.
- **Blocking objects** (pillars, crates). Interactable obstacles; slide along fixed trajectories to reshape the shadows cast by nearby light sources.
- **Exit gate.** A static trigger zone. Completing the level requires the Orb to enter this region while a valid shadow path connects start to exit.

2.4.4 User/Character

The player never directly controls the Orb's interaction with objects; all object manipulation is performed via mouse before or between Orb movements. The interaction model is:

1. Player uses the mouse to position light sources and blocking objects, observing how the live shadow pattern changes.
2. Player uses the keyboard to roll the Orb along the shadow path toward the exit.
3. If the Orb falls (entered lit void), it respawns at the level start; the player adjusts the configuration and tries again.
4. When the Orb reaches the exit gate, the level ends and the next level loads automatically.

The Reset button is always accessible and returns all objects and the Orb to their initial positions for that level.

2.4.5 Character/Character

There are no NPCs, enemies, or secondary characters. The Orb is the only dynamic entity in the scene; all interaction is between the player, the Orb, and the environment.

2.5 Performance and Scoring

2.5.1 State Variables

The following variables are tracked at runtime for each level:

- **Orb position.** The current world-space position of the Orb. Reset to the level start position on fall or player reset.
- **Lantern position** (one per lantern). The current world-space position of each portable lantern within its movement boundary.
- **Spotlight angle** (one per spotlight). The current rotation angle of each wall-mounted spotlight about its fixed mount point.

- **Blocking object stop index** (one per blocking object). The index of the currently occupied stop position along each object's fixed trajectory.
- **Level completion flag.** Set to true when the Orb enters the exit gate; triggers the level-complete transition.
- **Levels unlocked.** A persistent list of which puzzle levels have been completed, used to populate the level-select screen.

2.5.2 Feedback

- **Live shadow update.** The shadow boundary redraws continuously as the player drags any light source or blocking object, giving immediate visual confirmation of every change.
- **Object hover highlight.** Interactable objects are visually highlighted when the mouse cursor hovers over them, indicating they can be manipulated.
- **Orb fall.** When the Orb enters a lit gap it plays a brief fall animation and audio cue before respawning at the level start. No objects are moved.
- **Level complete.** Entering the exit gate triggers a short completion animation and a distinct audio cue, followed by an automatic transition to the next level.
- **Interaction sounds.** Distinct sound cues play for lantern drag start/end, spotlight rotation, and blocking object slide.

2.5.3 Performance and Progress Metrics

Umbra does not record scores, star ratings, move counts, or elapsed time. Performance is binary: a level is either complete or not. The only persistent metric is the set of levels the player has completed, which unlocks them on the level-select screen. There is no leaderboard, no in-game timer display, and no penalty for resetting or falling.

3 Implementation

3.1 Design Assumptions

3.1.1 Hardware

Umbra targets standard gaming PC hardware. The primary performance constraint is real-time dynamic shadow rendering via Unreal Engine 5's Lumen system.

	Minimum	**Recommended**
OS	Windows 10 (64-bit)	Windows 11 (64-bit)
CPU	Intel Core i5-8400 / AMD Ryzen 5 2600	Intel Core i7-12700 / AMD Ryzen 7 5800X
RAM	8 GB	16 GB
GPU	NVIDIA GTX 1070 / AMD RX 5700	NVIDIA RTX 3070 / AMD RX 6800 XT
Storage	4 GB available	8 GB available (SSD)

3.1.2 Software

- **Game engine.** Unreal Engine 5 (UE5.3 or later), using C++ and Blurprit as wrappers for gameplay logic and Lumen for real-time global illumination and dynamic shadow rendering.
- **Operating system.** Windows 10/11 (64-bit). Development and testing are performed exclusively on Windows.

- **Development IDE.** Visual Studio 2022 (Community or Professional) for any C++ extensions; UE5 editor for Blueprint scripting and level design.
- **Version control.** Git with GitHub for source management; Unreal's `.gitignore` template applied to exclude derived build artefacts.
- **Asset pipeline.** Blender for 3D modelling and UV unwrapping; Adobe Photoshop / GIMP for texture authoring; Audacity for audio editing.

3.1.3 Algorithms and Techniques

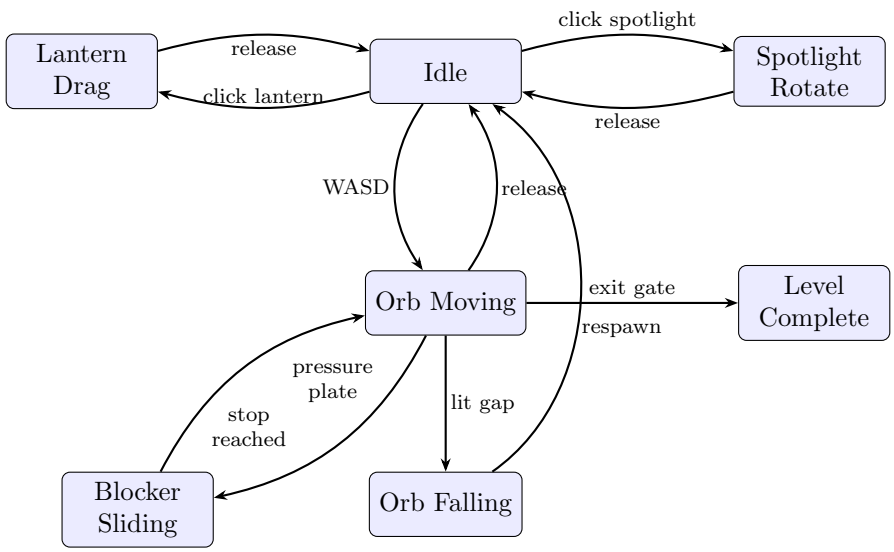
- **Real-time dynamic shadows (Lumen).** Unreal Engine 5's Lumen global illumination system computes shadow volumes continuously as light source positions and blocking object positions change. No baked lighting is used; all shadows are fully dynamic.
- **Shadow geometry as physical collision.** Shadow-covered regions are mapped to runtime collision volumes. When a light source or blocking object moves, affected collision volumes are updated every frame so the Orb's traversable surface always matches the visible shadow pattern.
- **Lantern drag (boundary-constrained free movement).** A mouse drag event translates the lantern's world position along the camera's view plane. A bounding volume defined per lantern at level-design time clamps the position, preventing the lantern from leaving its valid region.
- **Spotlight rotation (pivot-constrained rotation).** A mouse drag event computes the angle between the spotlight's mount point and the cursor's world-space projection, clamped to a designer-specified angular range.
- **Blocking object trajectory and snap.** Each blocking object follows a pre-defined spline or linear path. When the player interacts with a blocking object, it moves to the nearest valid stop position along its trajectory and snaps into place.
- **Orb physics.** The Orb uses UE5's built-in sphere physics with gravity and rolling friction. Movement is driven by keyboard input applying forces to the physics body; the Orb falls under gravity when it rolls off a shadow edge into lit void.

3.2 Storyboards

3.3 Design Logic

3.3.1 Finite State Machines — Events / States / Effects

The game's interaction layer is modelled as a finite state machine with seven states. **Idle** is the default resting state; all others are transient and return to Idle (or advance to Level Complete) on their terminal event. A **Reset** event is available in Idle, Orb Moving, Lantern Drag, and Spotlight Rotate states; it returns all objects and the Orb to their level-start positions and always transitions to Idle.



State descriptions:

- **Idle.** Default state. No input is being processed. The player may hover over interactable objects, press WASD, click a light source, or press Reset.
- **Lantern Drag.** Left mouse button is held on a portable lantern. The lantern follows the cursor world-space projection each frame, clamped to its movement boundary. Shadow geometry updates continuously. Released mouse button returns to Idle.
- **Spotlight Rotate.** Left mouse button is held on a wall-mounted spotlight. The spotlight angle is driven by the cursor's angular offset from the mount point each frame, clamped to the designer-specified range. Shadow geometry updates continuously. Released mouse button returns to Idle.
- **Orb Moving.** WASD is held; input forces are applied to the Orb's physics body each frame. Stepping on a pressure plate transitions to Blocker Sliding (then back). Entering a lit gap transitions to Orb Falling. Reaching the exit gate transitions to Level Complete.
- **Blocker Sliding.** A blocking object is animating along its trajectory toward the next stop position. Player input is suspended. When the object snaps to the stop, the machine returns to Orb Moving.
- **Orb Falling.** The Orb has entered lit void. A brief fall animation and audio cue play. All objects remain where the player left them. On animation end the Orb respawns at the level start and the machine returns to Idle.
- **Level Complete.** The Orb has entered the exit gate. A completion animation and audio cue play, then the next level loads (or the Victory screen if all levels are done).

3.3.2 User Solution / Actions

The following walkthrough traces a complete solution to the tutorial level (**The Antechamber**) and maps each player action to the corresponding FSM transitions.

1. **Level loads — Idle.** The Orb sits at the start position. A portable lantern rests near the centre of the platform, casting light away from a gap that blocks the path to the exit. A blocking

pillar stands to the side of the gap, and a pressure plate is visible beyond the first gap.

2. **Drag the lantern to bridge the first gap (Idle → Lantern Drag → Idle).** The player clicks and holds the lantern, then drags it toward the near edge of the gap. As the lantern moves, its shadow sweeps across the void, and the gap region gradually becomes physically solid. When the shadow fully spans the gap the player releases the mouse. The machine returns to Idle with the gap bridged by shadow.

3. **Roll the Orb across the shadow bridge (Idle → Orb Moving → Idle).** The player presses WASD to roll the Orb forward along the shadow-covered path. The Orb crosses the gap without falling because the shadow beneath it is solid. The player releases WASD after reaching the far platform; the machine returns to Idle.

4. **Trigger the pressure plate to slide the pillar (Idle → Orb Moving → Blocker Sliding → Orb Moving → Idle).** A second gap separates the far platform from the exit. The blocking pillar, if repositioned, would cast a shadow that spans it. The player presses WASD to roll the Orb onto the pressure plate marked on the floor. The Orb Moving state detects the pressure-plate overlap and transitions to Blocker Sliding: the pillar animates along its fixed trajectory to its second stop position, now blocking the light source from the correct angle. The shadow extends across the second gap. Once the pillar snaps to the stop the machine returns to Orb Moving. The player releases WASD; the machine returns to Idle.

5. **Roll the Orb to the exit gate (Idle → Orb Moving → Level Complete).** The player presses WASD to roll the Orb along the newly formed shadow path across the second gap and into the exit gate. The Orb Moving state detects the exit-gate overlap and transitions to Level Complete. The completion animation plays and the next level loads.

3.4 Software Versions

3.4.1 Alpha Version

Due: Wed. Mar. 18. The Alpha is a vertical slice demonstrating all core mechanics in a single fully playable level.

- **Core shadow mechanic.** Real-time dynamic shadows; shadow-covered regions are mapped to runtime collision volumes so the Orb can traverse them.
- **Lantern drag.** One portable lantern freely draggable within its boundary constraint; shadow updates continuously.
- **Blocking objects and pressure plates.** One or two blocking objects that slide to designer-specified stop positions when the Orb steps on the corresponding pressure plate.
- **Orb physics.** Keyboard-driven sphere with UE5 physics: gravity, rolling friction, and respawn on fall.
- **One playable level.** *Level 1 — The Collapsed Bridge*: a single self-contained puzzle using the lantern and two blocking pillars to span a central chasm.
- **Minimal UI.** Level name display and a Reset button (returns all objects and the Orb to start positions).

- **Placeholder art and audio.** Untextured geometry and placeholder sound cues sufficient for playtesting.

3.4.2 Beta Version

Due: Wed. Mar. 25. The Beta expands the Alpha to a complete multi-level experience with polished content and secondary mechanics.

- **Spotlight rotation.** Wall-mounted spotlights freely rotatable within their angular range; directed-cone shadow updates continuously.
- **All puzzle levels.** *Level 2 — The Torch Hall* (spotlights + blocking objects) and *Level 3 — The Inner Sanctum* (combined mechanics) added and playable end-to-end.
- **Tutorial level.** *The Antechamber* included with on-screen prompts (see §3.4.3).
- **Level-select screen.** Main menu listing all four spaces; completed levels are marked unlocked.
- **Victory screen.** Displayed after completing all three puzzle levels.
- **Polished art.** Textured Roman-ruin geometry, particle effects on light sources, and atmospheric skybox.
- **Polished audio.** Ambient music track, distinct sound cues for lantern drag, spotlight rotation, blocker slide, Orb fall, and level completion.
- **Hover highlights.** Interactable objects highlight on mouse-over to communicate affordance clearly.
- **Bug fixes and playtest feedback.** Issues identified in the Alpha playtest resolved; puzzle difficulty tuned.

3.4.3 Tutorials

The tutorial level (**The Antechamber**) is a short, guided introduction that teaches all three interaction mechanics in sequence with no dead ends and no prior knowledge required.

1. **Shadow mechanic introduction.** A single gap separates the Orb from a small platform. A lantern is already positioned so its shadow almost spans the gap. An on-screen prompt reads: *"Drag the lantern to cast a shadow bridge."* The player adjusts the lantern; the shadow snaps across the gap. A second prompt reads: *"Walk across the bridge."*
2. **Pressure-plate and blocking-object introduction.** The Orb arrives at a second gap. A pressure plate is visible on the current platform, and a blocking pillar stands nearby. An on-screen prompt reads: *"Step on the pressure plate to move the pillar."* The Orb rolls onto the plate; the pillar slides to its second stop, extending the lantern's shadow across the second gap. The player rolls the Orb across to complete the level.
3. **Respawn demonstration (optional).** If the player rolls the Orb into a void before completing either step, the fall animation plays and the Orb respawns at the level start with all objects in their current positions. No explicit prompt is shown; the mechanic is self-evident.

On-screen prompts appear as brief text overlays anchored to the relevant object and disappear automatically once the player performs the indicated action. No prompts appear in the three main puzzle levels; the tutorial is the only guided section of the game.

4 Work Plan

4.1 Tasks

CC = Cecilia Chen JP = Jacky Park YT = Yiding Tian

	Task	Time (days)	Member
1	Set up base project (UE5, Git, asset folders)	3	All
2	**Shadow and Light System**		
2.1	Lumen real-time shadow configuration	5	CC
2.2	Shadow-to-collision volume mapping	7	CC, JP
2.3	Lantern boundary constraint system	3	JP
2.4	Spotlight angular constraint system	3	JP
3	**Player Interaction**		
3.1	Lantern drag (mouse-to-world, boundary clamp)	4	JP
3.2	Spotlight rotation (cursor angle, range clamp)	3	JP
3.3	Pressure plate trigger	2	JP
3.4	Blocking object trajectory and snap	4	JP
3.5	Object hover highlight	2	JP
4	**Orb Character**		
4.1	Sphere physics setup	3	CC
4.2	WASD keyboard input	2	CC
4.3	Fall detection and respawn	2	CC
5	**Level Design**		
5.1	Tutorial — The Antechamber	4	All
5.2	Level 1 — The Collapsed Bridge	5	All
5.3	Level 2 — The Torch Hall	5	All
5.4	Level 3 — The Inner Sanctum	7	All
6	**UI and HUD**		
6.1	Main menu and level-select screen	4	YT
6.2	In-game HUD and Reset button	2	YT
6.3	Level-complete and Victory screens	2	YT
6.4	Tutorial on-screen prompts	2	YT
7	**Art and Assets**		
7.1	Orb model and texture	3	CC
7.2	Environment geometry (platforms, ruins)	7	YT
7.3	Light source models (lantern, spotlight)	3	YT
7.4	Blocking object models (pillar, crate)	3	YT
7.5	Particle effects and skybox	3	YT
8	**Audio**		
8.1	Ambient background music	3	YT
8.2	Sound effects	3	YT
9	Testing	7	All
10	Polishing and extra features		All

4.2 Milestones

4.2.1 Minor Milestones

Date	Milestone
Wed. Mar. 4	UE5 project initialized; Git repo set up; all members able to build and run
Sat. Mar. 7	Real-time Lumen shadows working: moving a light source visibly reshapes the cast shadow
Tue. Mar. 10	Shadow-to-collision mapping working: Orb can traverse shadow-covered regions and falls into lit void
Thu. Mar. 12	Orb physics complete: WASD movement, gravity, rolling friction, fall detection, and respawn
Sat. Mar. 14	Lantern drag with boundary constraint working; pressure plate trigger and blocker snap working
Mon. Mar. 16	Level 1 full loop playable: lantern, two blockers, exit gate, Reset button
Fri. Mar. 20	Spotlight rotation working; all four level layouts blocked out
Mon. Mar. 23	Tutorial level with on-screen prompts complete; level-select screen and all UI screens implemented
Thu. Mar. 26	Full art pass complete: all meshes textured, particle effects and skybox in place
Sat. Mar. 28	Audio complete: ambient music and all sound effects integrated
Sun. Mar. 29	Final bug fixes, puzzle tuning, and performance pass complete

4.2.2 Major Milestones

Date	Milestone
Mon. Mar. 2	Game Design Document submitted
Wed. Mar. 18	Alpha Version / Playtest: one fully playable level with core shadow mechanic
Wed. Mar. 25	Beta Version / Playtest: all four levels, polished art and audio, complete UI
Mon. Mar. 30	Final Version / Demo: release build packaged and submitted
Wed. Apr. 1	Game Trailer and Gameplay Videos submitted

4.3 Development Schedule

	A	B	C	D	E	F
1	Task Name	Week 1 (2-Mar)	Week 2 (9-Mar)	Week 3 (16-Mar)	Week 4 (23-Mar)	Week 5 (30-Mar)
2	Dates	2-Mar	9-Mar	16-Mar	23-Mar	30-Mar
3	Design Doc	All				
4	Project Setup & Scaffolding	CC & YT	JP			
5	Shadow & Light System	CC & JP	JP	CC & JP		
6	Player Interaction		JP	JP		
7	Orb Character	CC		CC		
8	Level Design		JP	All	YT	
9	UI & HUD			YT	YT	
10	Alpha Build & Integration			*1 All		
11	Art & Assets			CC & YT	YT	
12	Audio				YT	
13	Beta Build & Integration				*2 All	
14	Testing & Final Demo				All	*3 All
15						
16	*1 = Alpha Playtest (Mar 18)					
17	*2 = Beta Playtest (Mar 25)					
18	*3 = Final Demo (Mar 30) / Videos (Apr 1)					
19	CC = Cecilia Chen [out Mar 6-15] | JP = Jacky Park | YT = Yiding Tian [out Mar 7-14]					